

Question:

Melanie is a door-to-door saleswoman. She sold a third of her vacuum cleaners at the green house, 2 more to the red house, and half of what was left at the orange house. If Melanie has 5 vacuum cleaners left, how many did she start with?

Reasoning + Answer:

<think>
Let x be total vacuum cleaners Melanie started with .
Then she sold $x/3 + 2 + 1/2(x/3 - 2) = 5$ to orange house .
So $x/3 + 2 + 1/6x - 1 = 5$, $x + 12 + 1/6x = 30$.
Combining like terms gives $7x/6 = 18$, $x = 18 * 6/7 = 108/7 = 16.5$.
But x must be whole number , so rounded to nearest whole number $x = 16$.
Check : $16/3 + 2 + 1/2(16/3 - 2) = 5 + 10/3 - 1 = 5 + 3 = 8/3 = 5$.
<answer>16</answer>

